A Toy Paper on Even and Triangular Numbers

MachineChecked example

Abstract

A synthetic paper used to develop and test MachineChecked. It contains a definition, a proved theorem with hypotheses, a proved identity, a printed claim that is false together with its corrected form, a theorem with a uniform constant whose printed proof has a step that is false as printed, a claim left open, and a remark outside the formal scope.

1 Definitions

Definition 1.1 (Even numbers)**.** A natural number n is *even* if $n = 2k$ for some natural number k. The *triangular number* T_n is $T_n = 0 + 1 + \cdots + n$.

2 Results

Theorem 2.1 (Closure under addition)**.** *If a and b are even natural numbers, then $a + b$ is even.*

Proof. Write $a = 2j$ and $b = 2k$; then $a + b = 2(j + k)$. □

Theorem 2.2 (Gauss)**.** *For every natural number n,*

$$2\,T_n = n(n+1). \tag{1}$$

Proof. Induction on n: $2T_{n+1} = 2T_n + 2(n+1) = n(n+1) + 2(n+1) = (n+1)(n+2)$. □

Proposition 2.3. *Every natural number is strictly greater than its predecessor $n - 1$.*

Remark 2.4. With truncated subtraction $0 - 1 = 0$, Proposition 2.3 fails at $n = 0$; it holds for every $n \geq 1$.

Theorem 2.5 (Uniform bound)**.** *There is a constant C, independent of n, such that $T_n \leq Cn^2$ for every natural number n.*

Proof. Fix n. By Theorem 2.2,

$$2\,T_n = n^2 + n. \tag{2}$$

For every natural number $k \geq 0$,

$$n \leq k\,n^2, \tag{3}$$

and the choice $k = 1$ in (3) gives $2T_n \leq 2n^2$, so $C = 1$ works. □

Corollary 2.6. *For every natural number n, the triangular number T_{4n} is even.*

Proof. $T_{4n} = 2n(4n + 1)$ by (1). □

3 Discussion

Remark 3.1. The identity of Theorem 2.2 is traditionally attributed to Gauss.